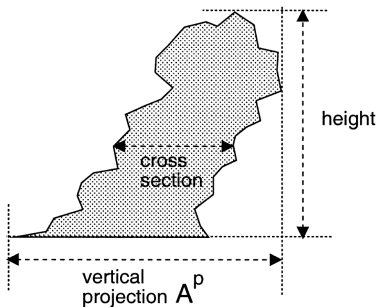


Large-eddy simulations (LES) of shallow cumulus. Each cloud gets one size; the size histogram shows which clouds dominate cloud cover and vertical transport.



$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

Three LES cases

BOMEX: ocean, trade winds.

SCMS: land, Florida.

ARM: land, diurnal cycle.

Symbols

cloud connected set of saturated grid cells

A^p vertically projected area of one cloud: its footprint seen from above

l cloud size, m

$\mathcal{N}(l)$ number of clouds per unit size, m^{-1}

N total number of clouds in the domain

Find each cloud in 3D $\rightarrow$ project it vertically $\rightarrow$ histogram of the sizes l

A power law below a variable size break

2

Small clouds are by far the most numerous. Below a size ℓ_d their number falls as a power law of size; above ℓ_d it falls faster.

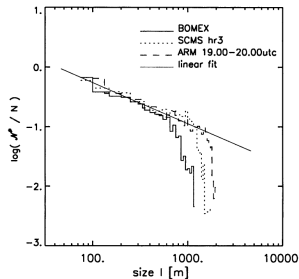


Fig. 4a: size density against l .

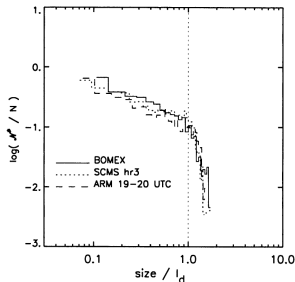


Fig. 6: against l/ℓ_d , the cases collapse.

$$\mathcal{N}(l) = al^b$$

$$b = -1.70 \text{ for } l < \ell_d$$

Symbols

ℓ_d scale break: size where the exponent changes. BOMEX 700 m, SCMS 1050 m, ARM 400 $\rightarrow$ 1250 m in the afternoon

a, b fit constants; b is the exponent
 $\mathcal{N}^* = l \ln(10) \mathcal{N}$, clouds per decade of size (plotted), slope $b + 1 = -0.70$

Same exponent in all cases: only the break ℓ_d changes

Weighting every cloud by its area gives the cloud cover contributed by each size. It grows up to ℓ_d and drops beyond, so mid-size clouds cover most of the sky.

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y} \propto l^{b+2} = l^{0.30} \quad (l < \ell_d)$$

$$a^p = \int_0^\infty \alpha^p(l) dl$$

Symbols

$\alpha^p(l)$ cloud cover from clouds of size l , per unit size, m^{-1}

a^p projected cloud cover: fraction of the domain covered, seen from above

$L_x L_y$ horizontal area of the domain

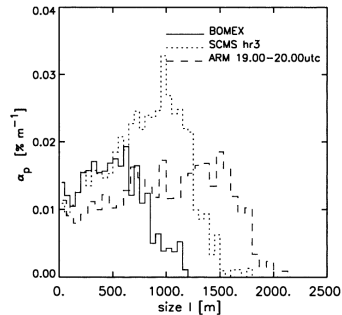


Fig. 7a: $\alpha^p(l)$, three cases.

Cloud cover peaks near the break ℓ_d

Mass flux peaks at larger sizes than cover

4

Mass flux is cloud cover times vertical velocity. Larger clouds rise faster, so transport is dominated by somewhat larger clouds; the smallest carry almost nothing.

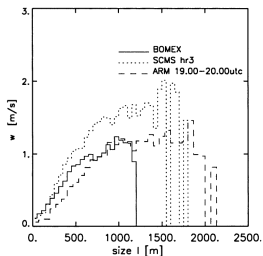


Fig. 7b: mean vertical velocity w .

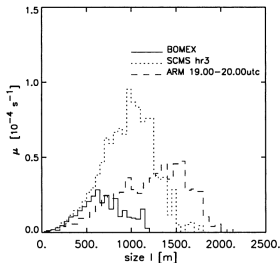


Fig. 7c: mass flux $\mu(l)$.

$$\mu(l, z) = \alpha(l, z) w(l, z)$$

$$\mu(l) = \frac{1}{h_c} \int_{h_c} \mu(l, z) dz$$

Symbols

$w(l, z)$ mean vertical velocity of clouds of size l at height z , m s^{-1}

$\alpha(l, z)$ their cloud cover at height z , per unit size

h_c depth of the cloud layer

μ mass flux divided by air density, per unit size, s^{-1}

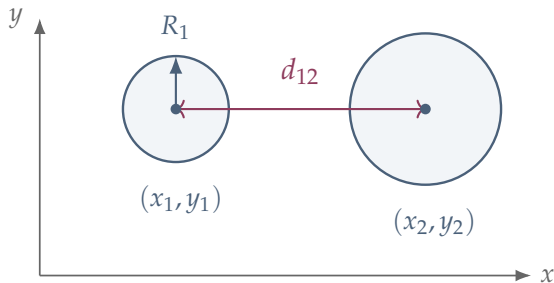
Mass flux peaks at larger cloud sizes than projected cover

No 2D distribution of thermals

5

Our question: how are thermals arranged in a horizontal plane? The paper measures clouds seen from above, a different object.

Spatial quantities (schematic)



Sought, not in the paper

R_i radius of thermal i at a fixed height

(x_i, y_i) position of its centre

d_{ij} distance between thermals i and j

Why cloud sizes do not answer it

A cloud is saturated air; a thermal can stay dry and never form a cloud.

Projection merges all heights and widens with tilt and shear.

Projected cumulus size does not determine thermal size or spacing

Scope: §3–5 and Appendix. The sketch defines the quantities sought. It represents no observed or simulated thermal field.